

Bank Customer Churn Analysis – Exploratory Data Analysis (EDA)

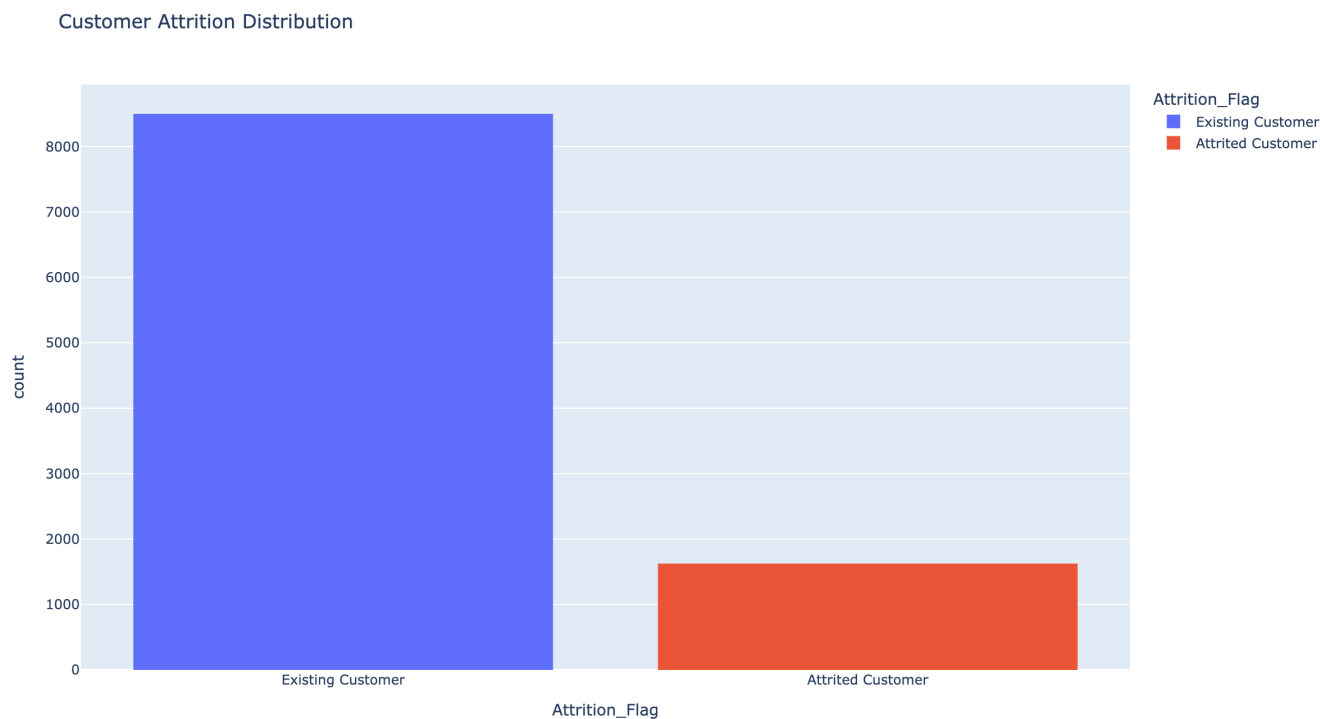
This project explores **customer churn behavior** using the Bank Churners dataset. The goal is to identify key factors that influence customer attrition and extract actionable business insights.

Importing Libraries

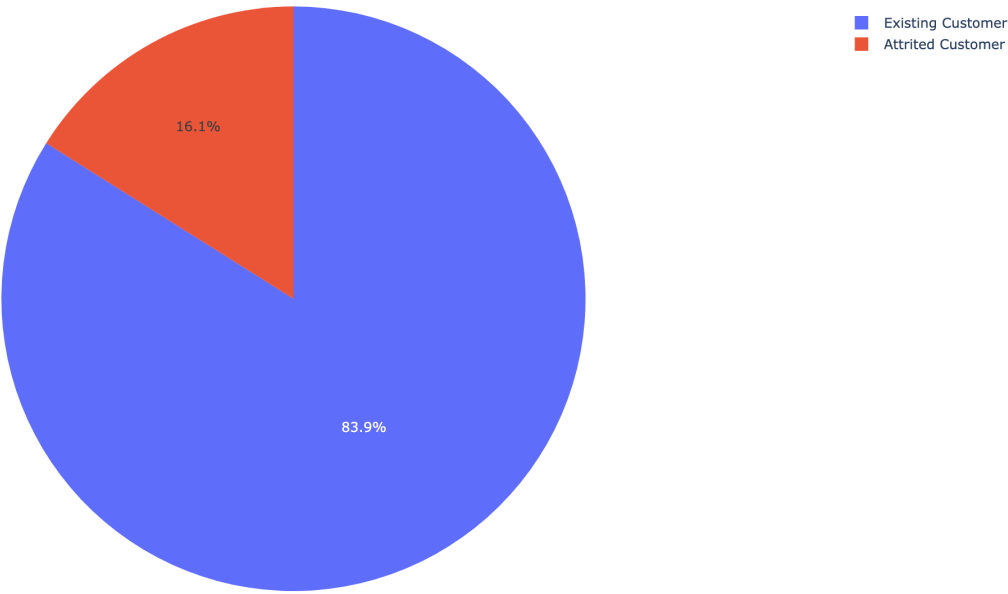
No missing values were detected in the dataset.

Data Cleansing

Target Variable Analysis (Churn Distribution)



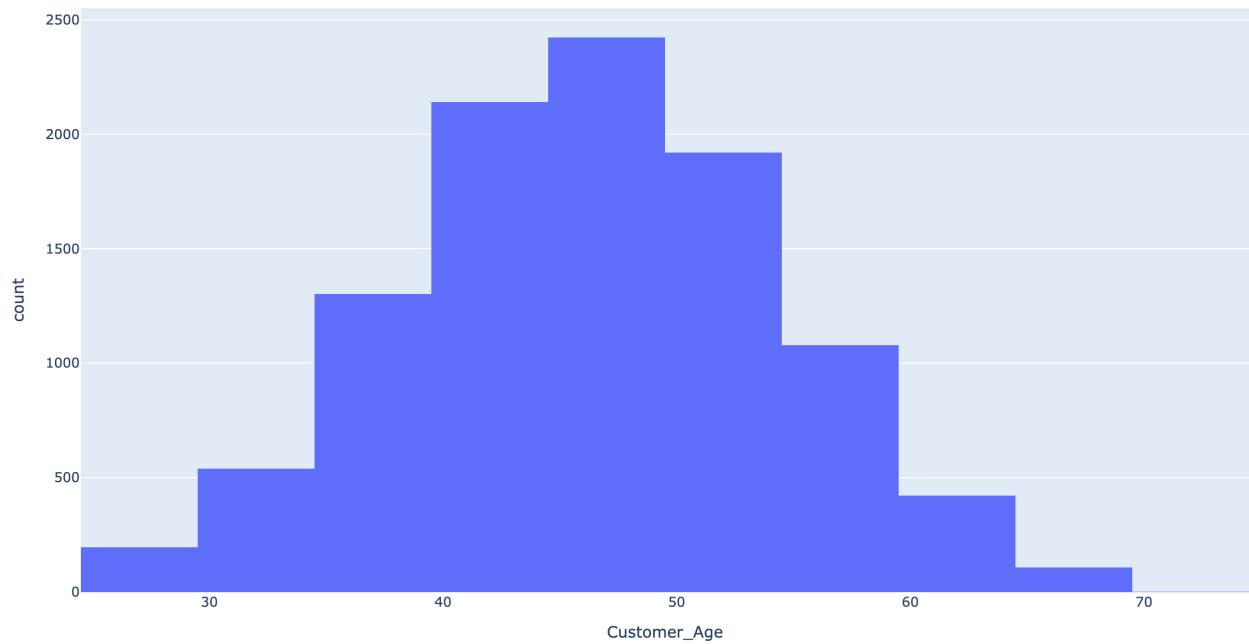
Customer Attrition Percentage(%)

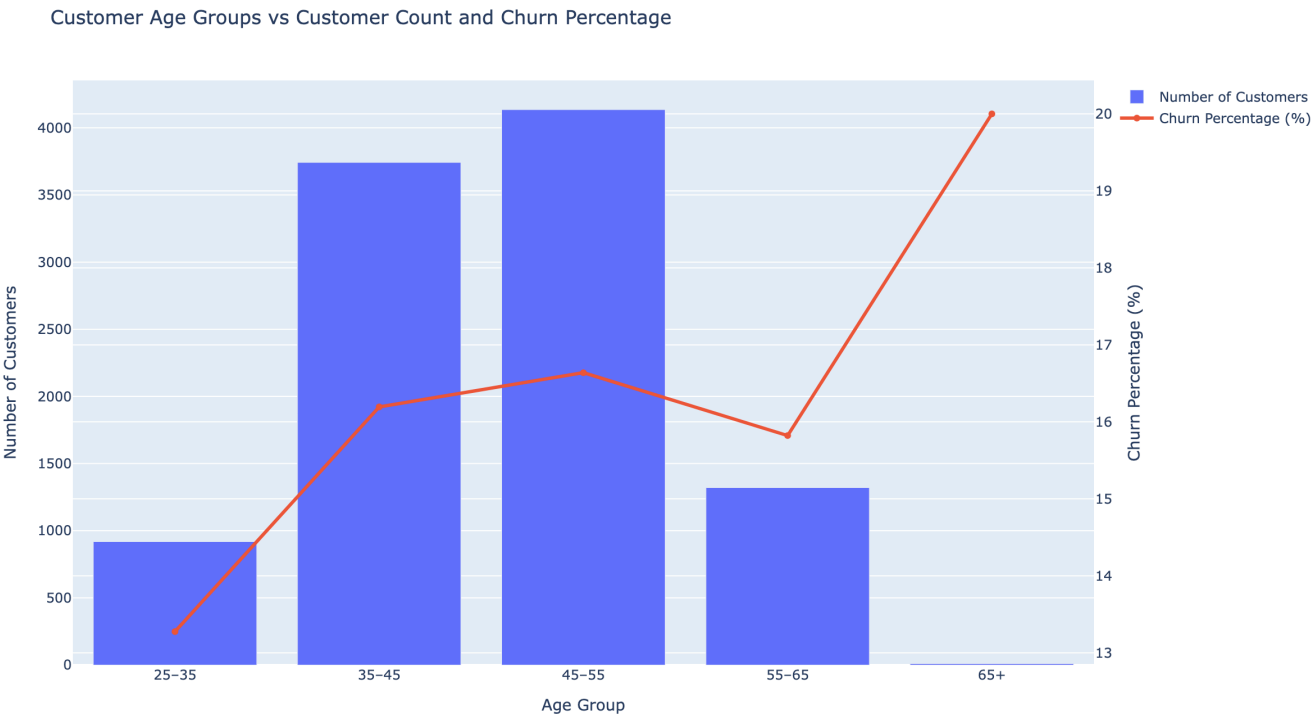


Only a small percentage of customers have churned, about 16%.

Age Distribution and Customer Age Groups vs Customer Count and Churn Percentage

Distribution of Customer Age

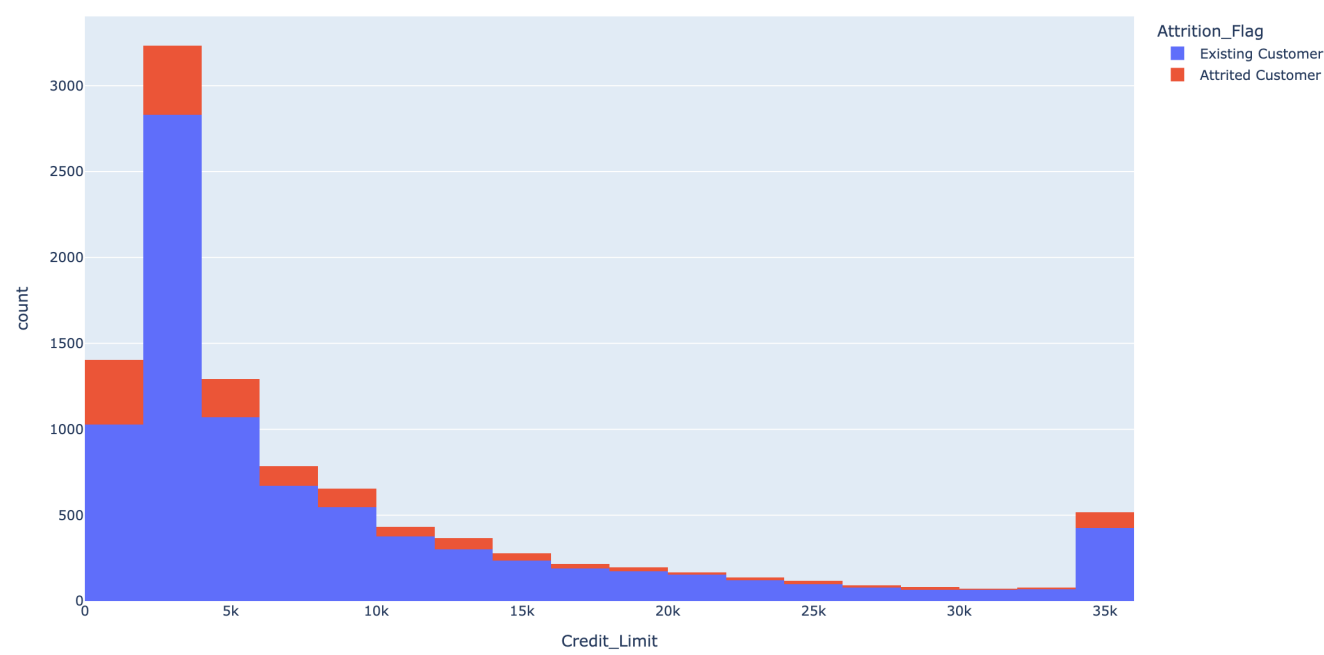




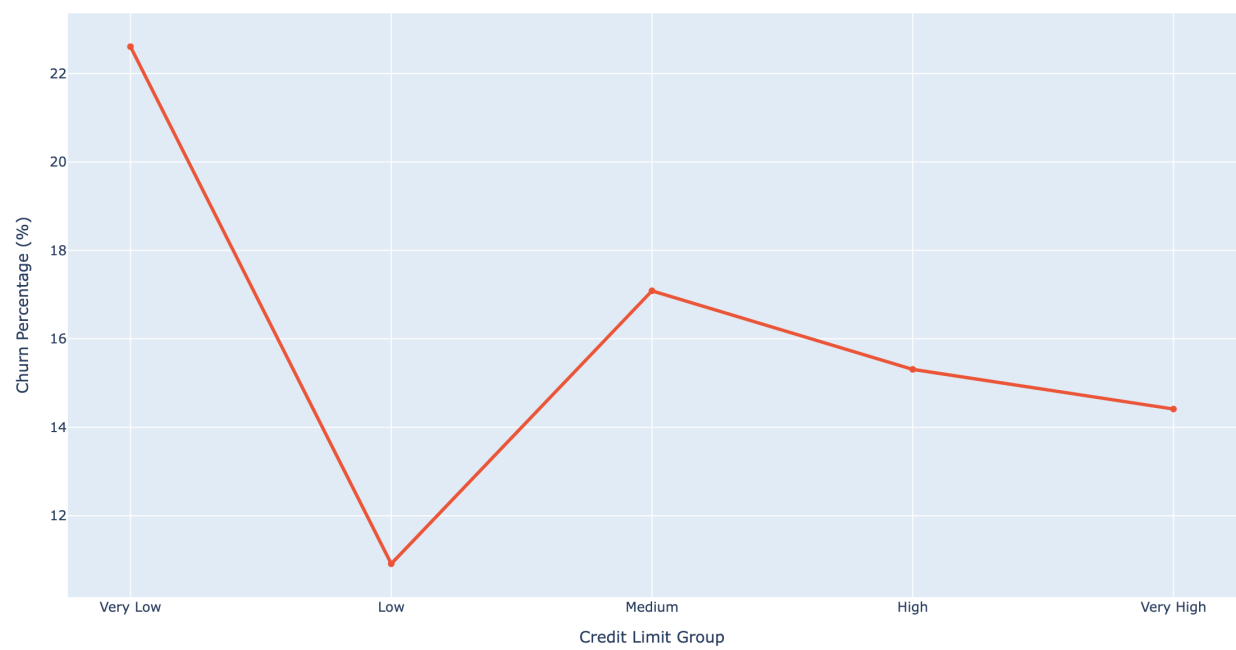
Younger customer segments tend to show relatively higher churn rates compared to older customers , suggesting that customer maturity and longer-term financial engagement may contribute to stronger retention behavior.

Credit Limit Distribution and Churn Percentage VS Credit Limit Groups

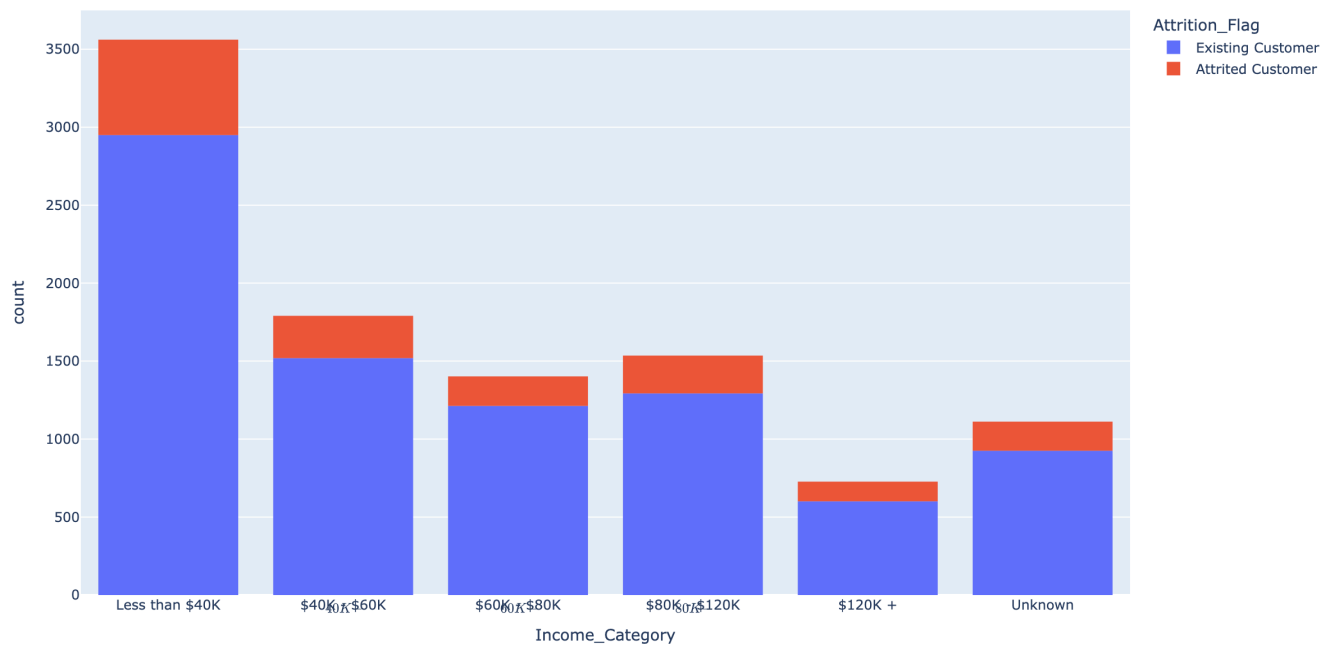
Distribution of Credit Limit



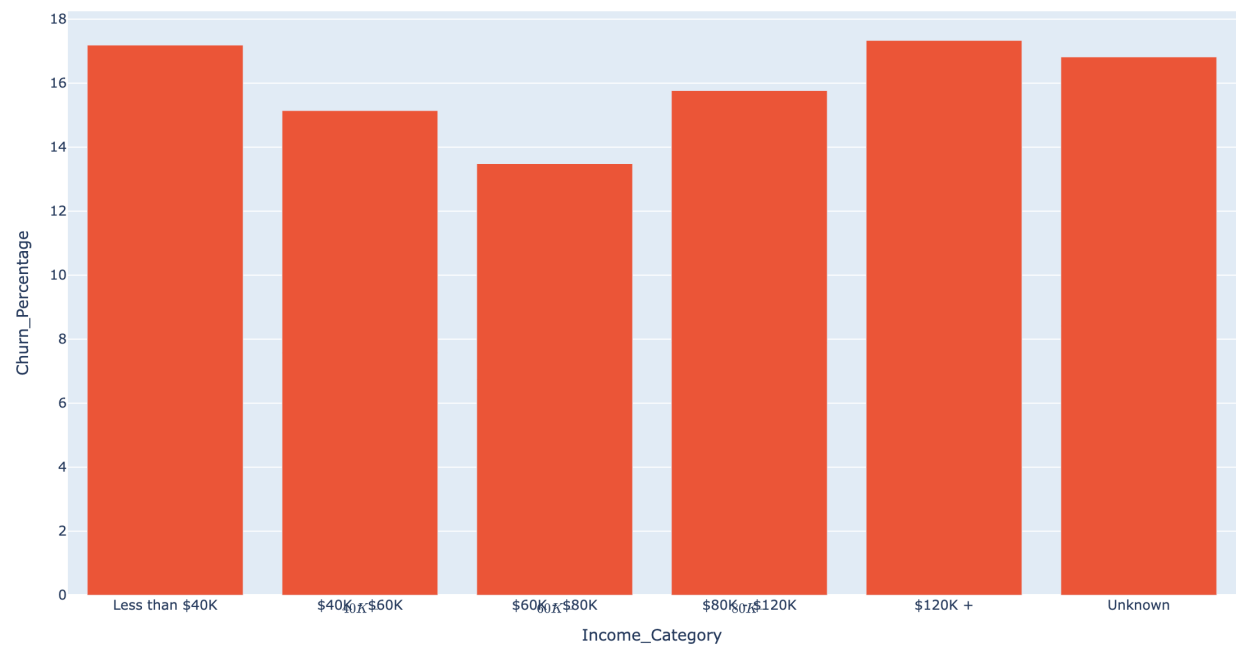
Churn % VS Credit Limit Group



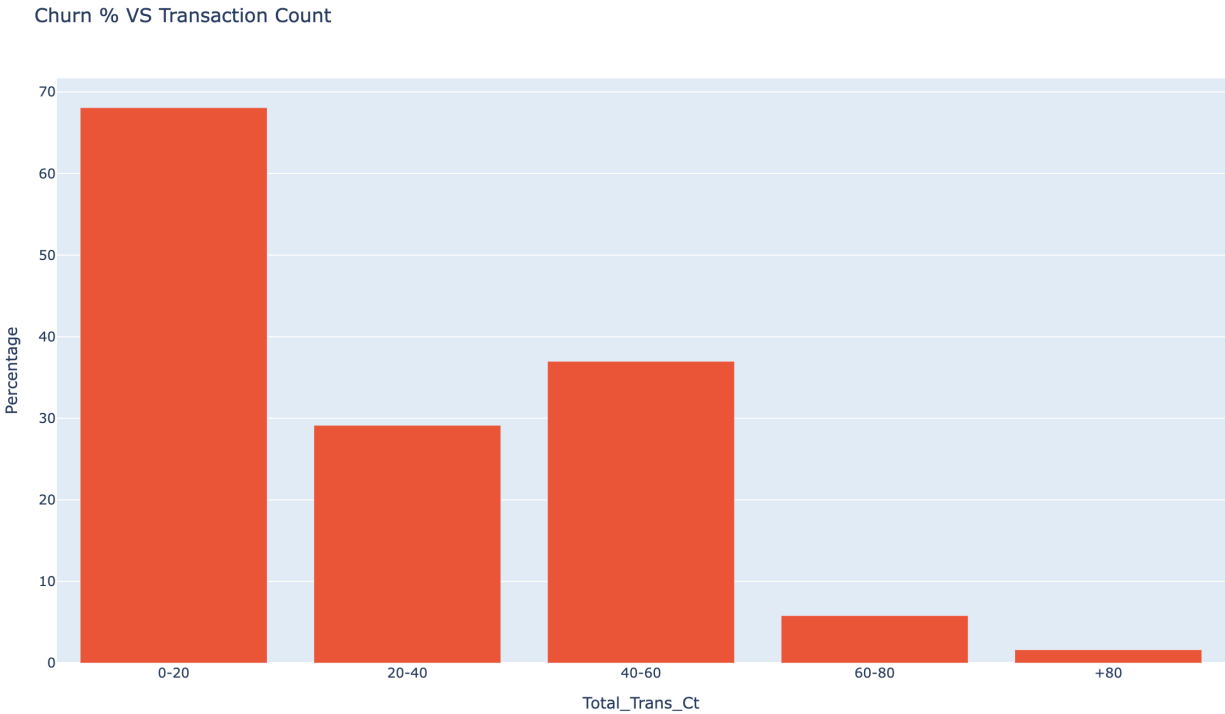
Income Category Distribution



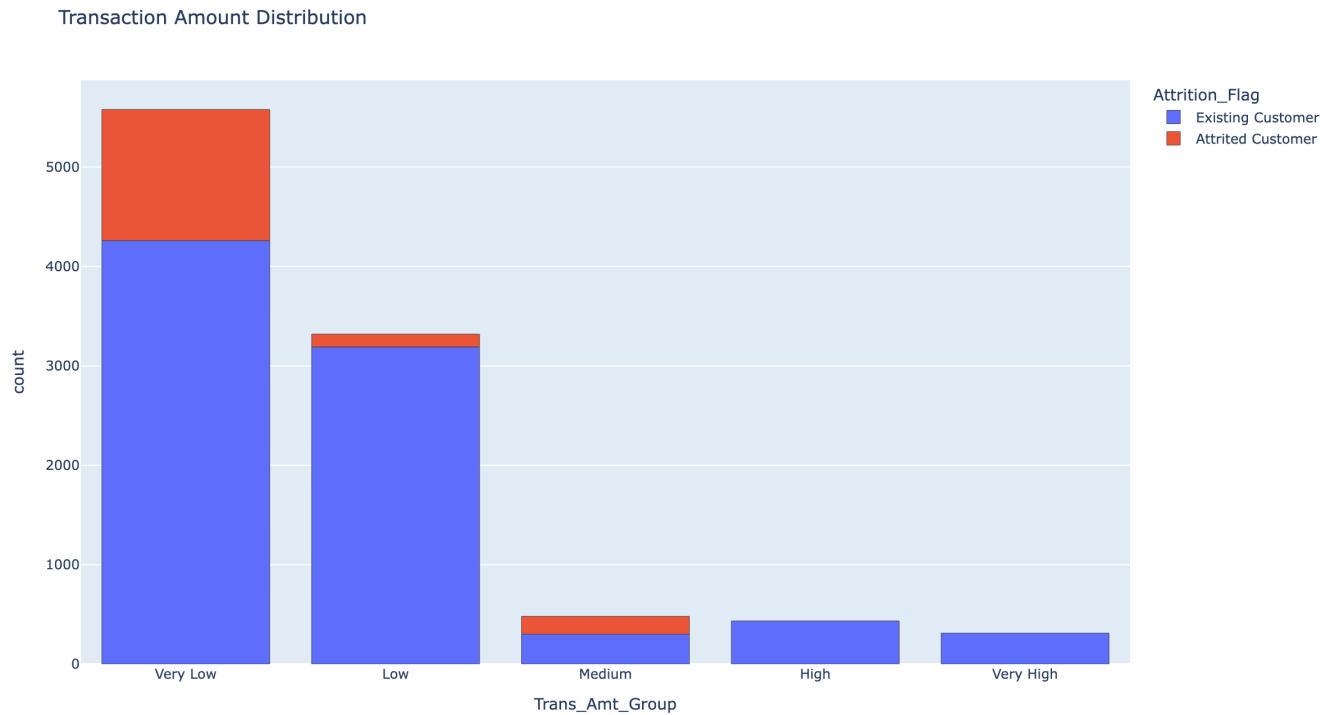
Churn % VS Income Category

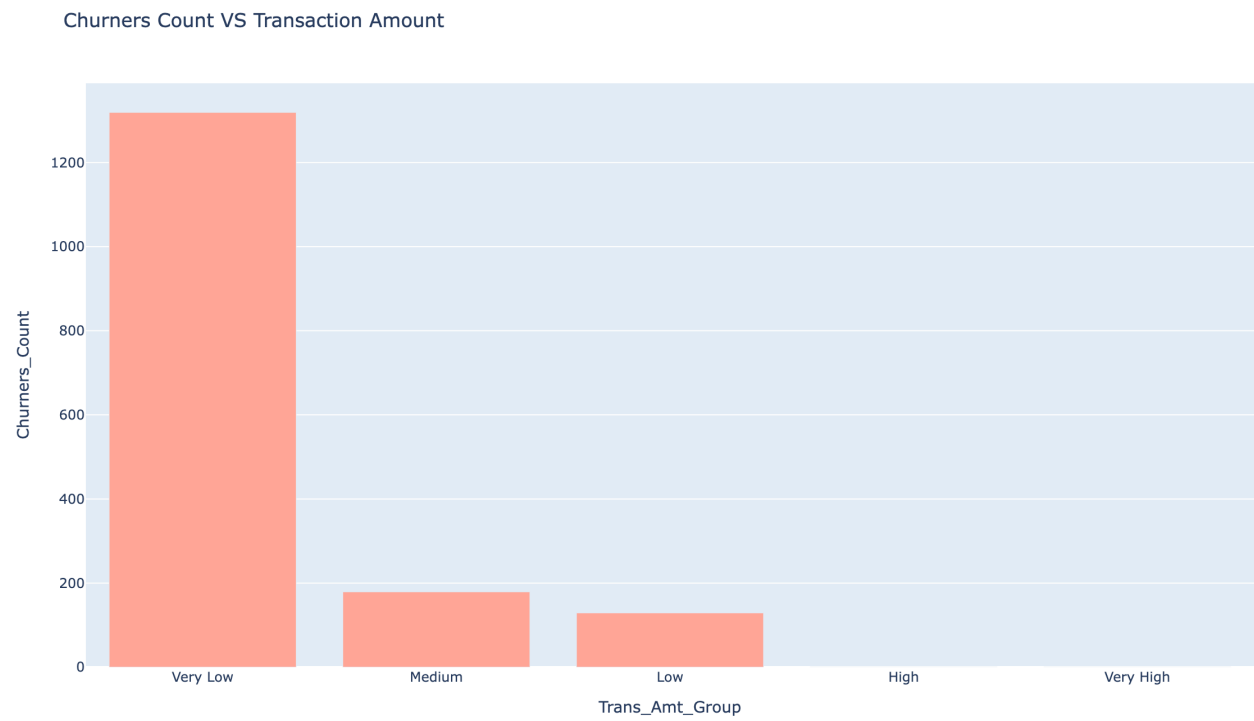
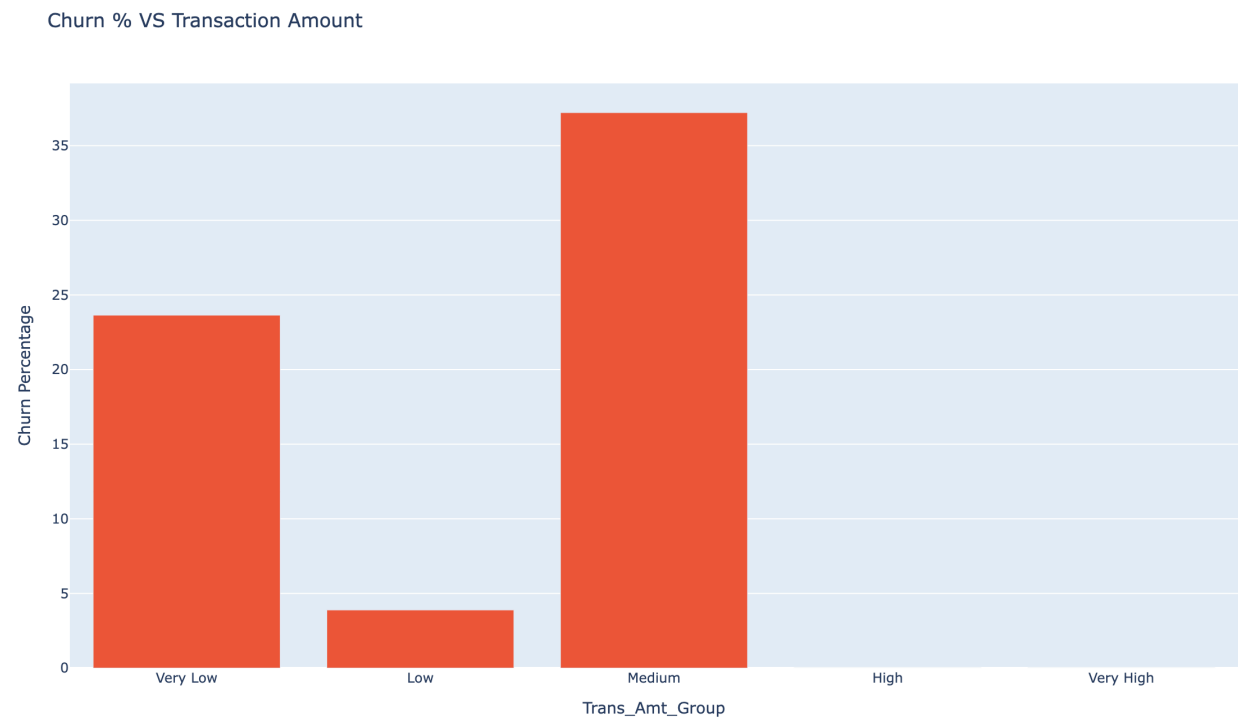


Churn Percentage VS Transactions Count



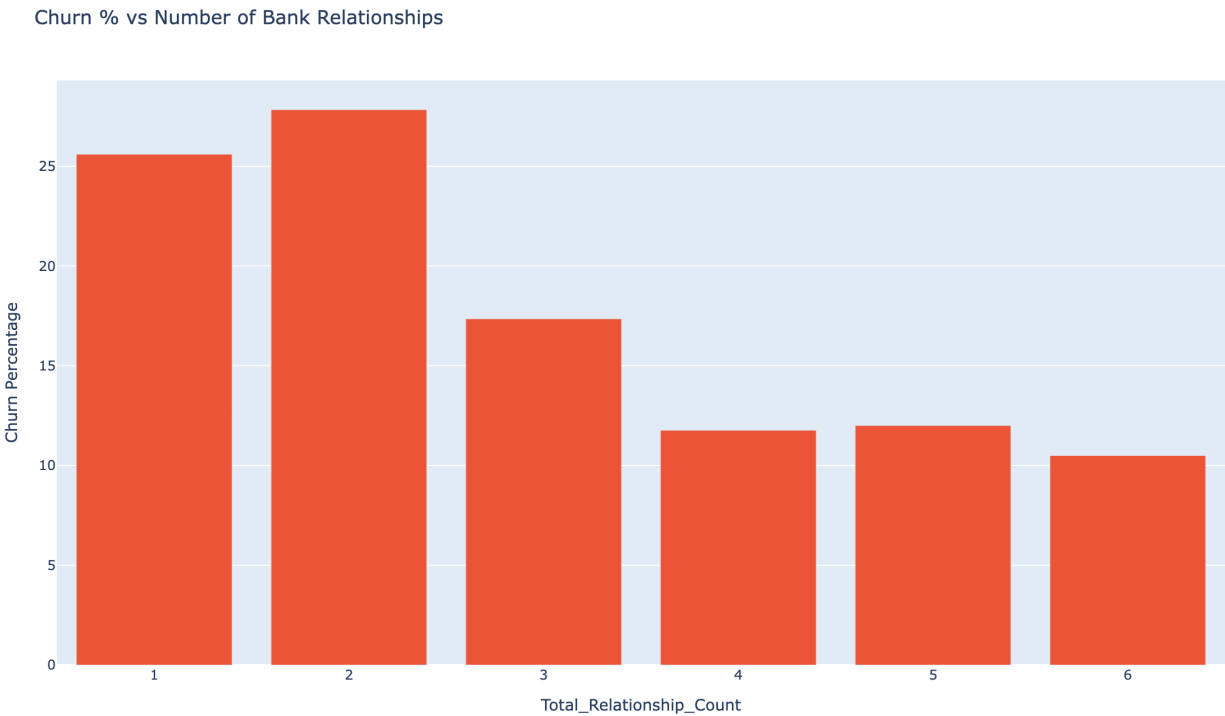
Transaction Amount Distribution and Churn Percentage VS Transactions Amount Group



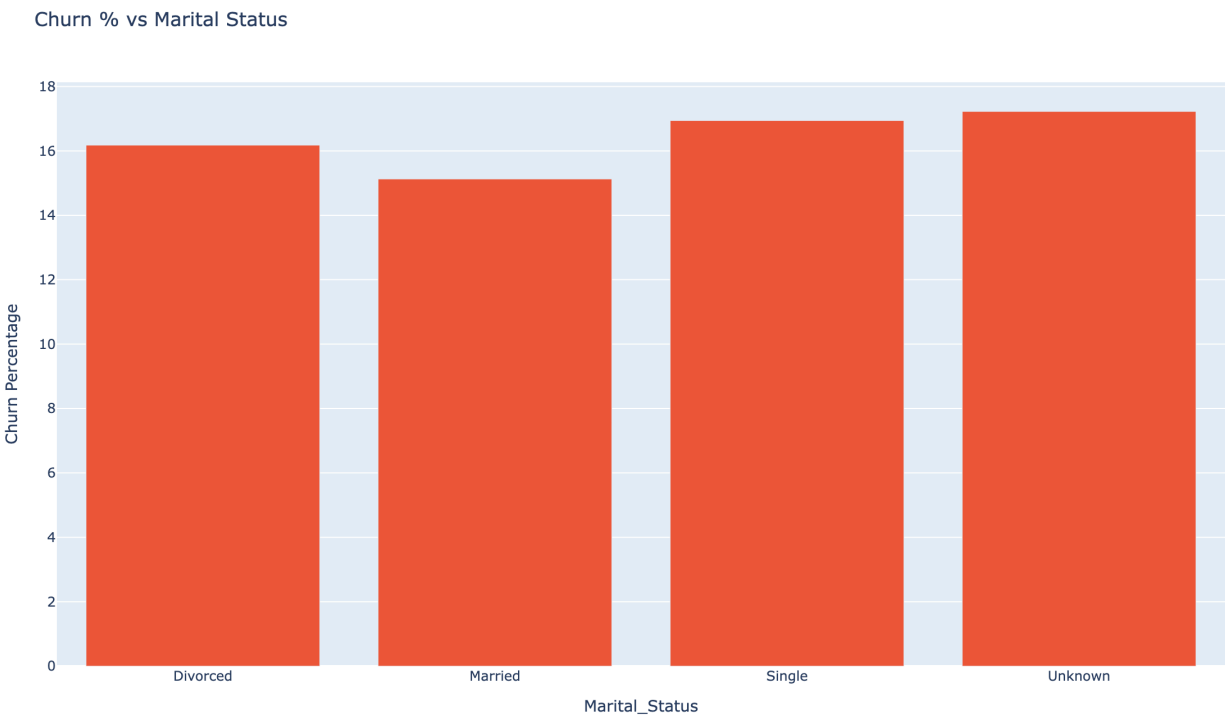


Customers in the **Medium transaction amount group** have the **highest churn rate**, indicating a higher likelihood of attrition compared to other groups.

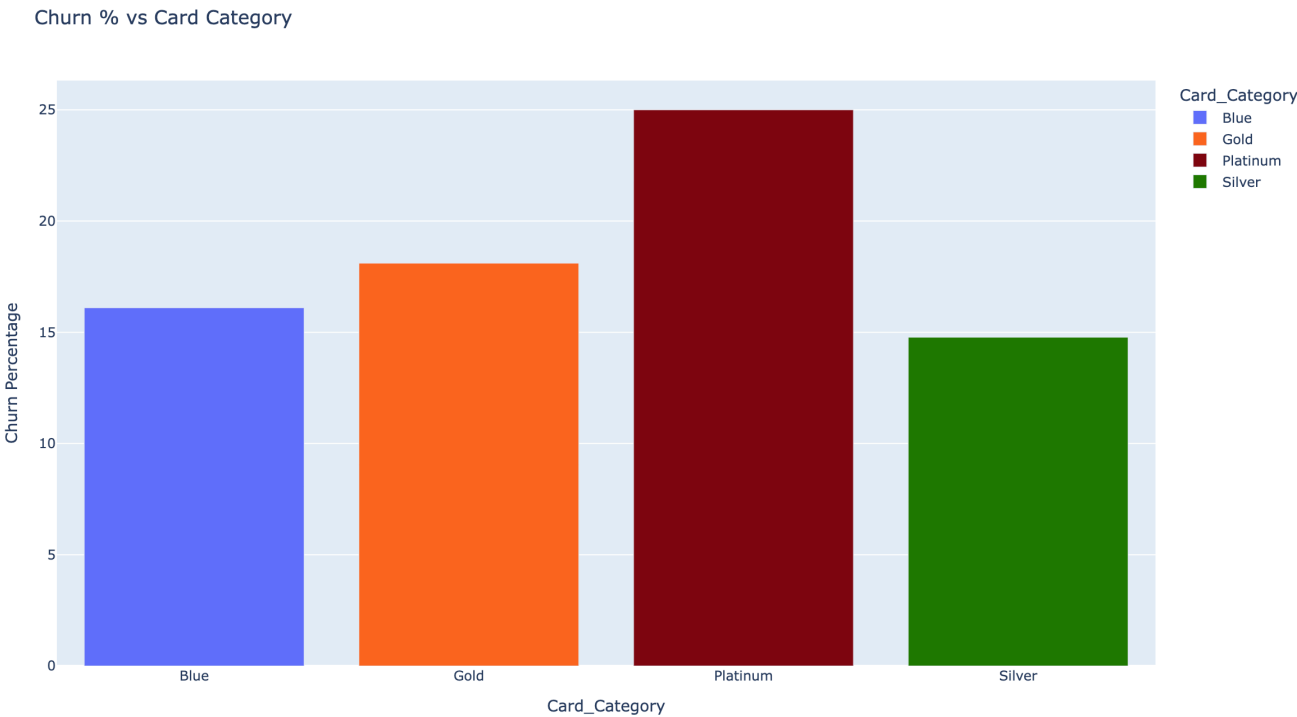
Churn Percentage VS Number of Bank Relationships



Customers with **fewer banking relationships** show significantly **higher churn rates** compared to customers with **multiple relationships**. This suggests that increasing cross-selling opportunities may improve customer retention.



Churn Percentage VS Card Category



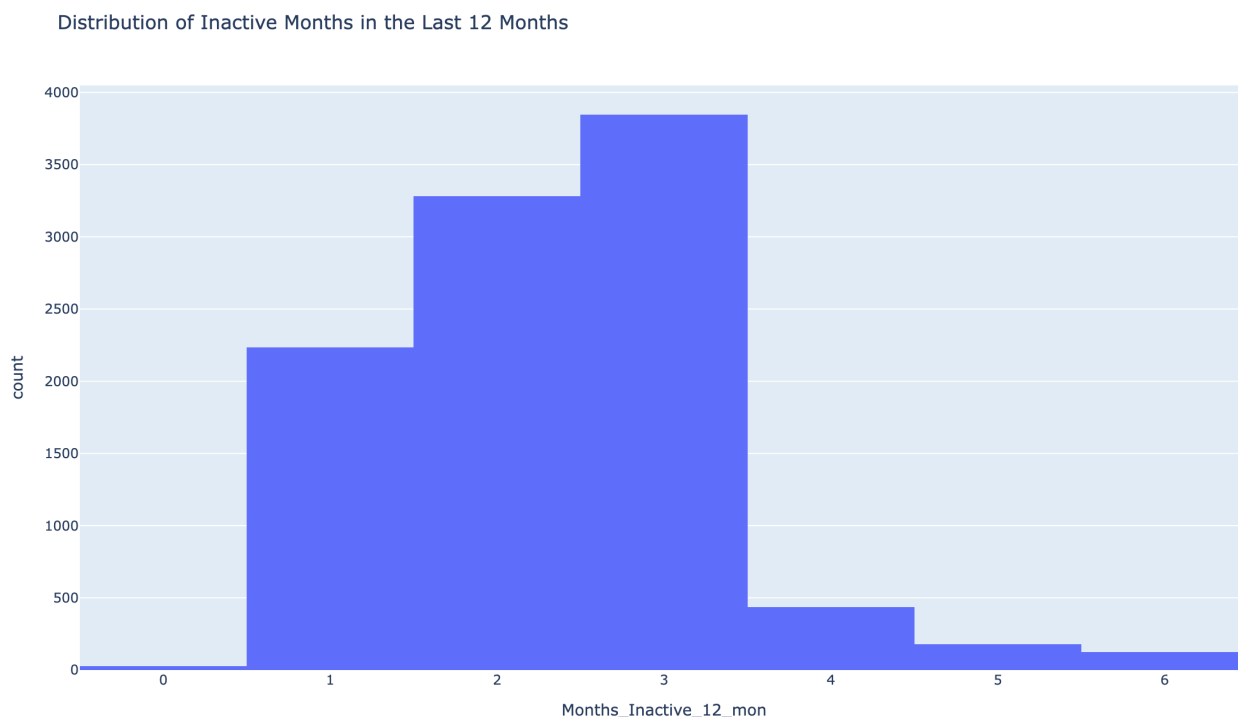
Customers holding **Platinum** cards exhibit the **highest churn rate among all card categories**, indicating a potential retention risk within this premium segment.

This may suggest unmet expectations regarding service value, benefits, or engagement compared to other card tiers.

In contrast, **Silver** and **Blue** cardholders show relatively **lower churn rates**, reflecting more stable retention behavior.

These findings highlight the importance of reviewing benefit structures and engagement strategies specifically for high-tier cardholders to improve customer loyalty.

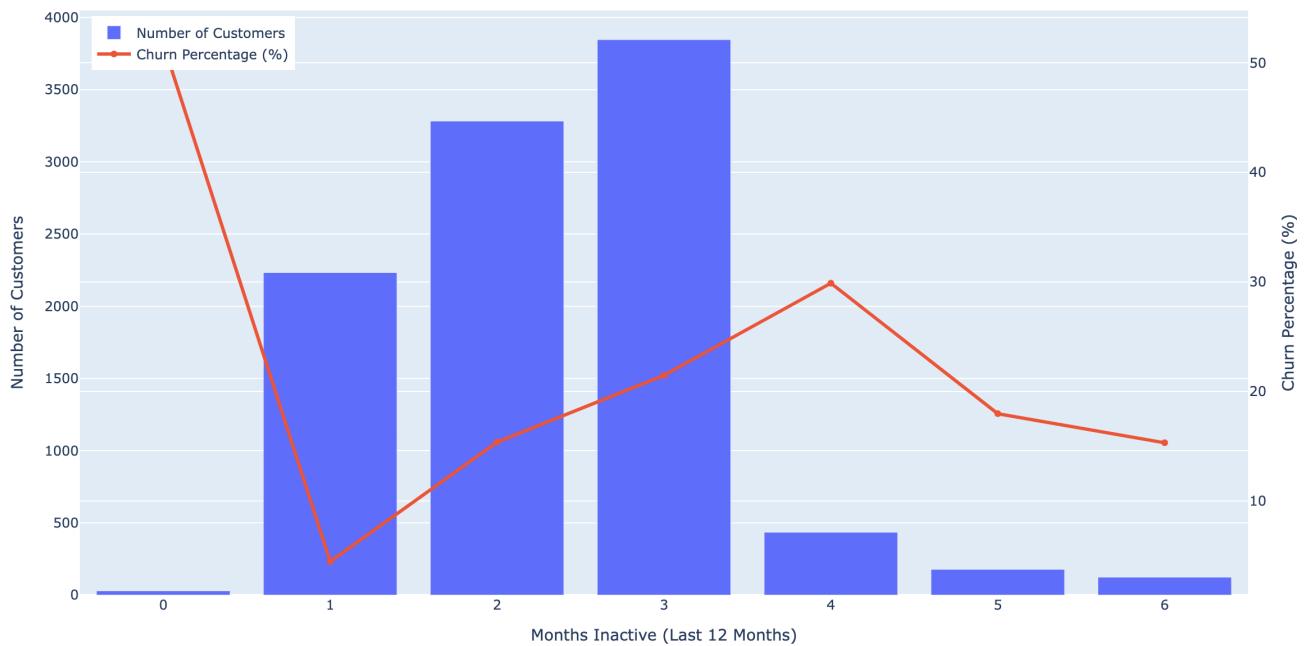
Inactive 12 Months Distribution and Inactive Months vs Customer Count and Churn Percentage



The majority of customers fall within 2–3 inactive months , while extended inactivity (4+ months) is relatively rare and may signal potential disengagement risk.

Although most customers remain inactive for only 2–3 months , higher inactivity durations should be monitored closely as they may represent early warning signals of potential customer attrition.

Inactive Months vs Customer Count and Churn Percentage



Although customers with zero inactive months show a relatively high churn percentage, the small sample size in this group reduces reliability.

Churn probability increases noticeably after 3 months of inactivity, indicating that extended inactivity is a strong early warning signal of potential customer attrition and should trigger proactive retention strategies.